COMPUTER VISION FOR THE AUTOMATED DETECTION OF POTHOLES AND GRATES IN URBAN ENVIRONMENTS

Subject: Artifical Intelligence **Teacher:** Felix Estanislao Vargas Quispe

Student: Orellana Cusihuaman Luis Anthony, Montufar Diaz Alfredo Gerardo, Vilca Ramos Luis Gerardo

ABSTRACT

The deterioration of urban road infrastructure poses a constant risk to traffic safety and generates significant maintenance costs. This project proposes an automated computer vision system capable of detecting and localizing potholes in road images by combining supervised learning techniques, object detection methods, and transformer-based architectures. The proposed approach aims to overcome the limitations of traditional manual inspections by providing a scalable and efficient tool for preventive urban infrastructure maintenance.

OBJECTIVES

- Automate the detection of potholes and grates in road images using computer vision.
- Compare the performance of classical machine learning models, deep learning approaches, and transformer-based architectures.
- Evaluate the models using standard object detection metrics.
- Support drivers and transportation operators through an early warning system for road hazards.

DATASET

Pothole Detection Dataset: YOLOv11 Optimized

3,900+ Road Images for Pothole Detection in YOLO11 Format

- Pothole Detection Dataset — Kaggle
- 3,940 imágenes / formato YOLO
- Splits: 3,345 train / 397 valid / 198 test

MODELS USED

RANDOM FOREST — SCIKIT-LEARN

Random Forest is a supervised machine learning algorithm based on an ensemble of decision trees. It combines the predictions of multiple trees to improve accuracy and reduce overfitting, making it effective for both classification and regression tasks. In this project, Random Forest serves as a classical machine learning baseline for detecting road surface anomalies.

YOLO11N — ULTRALYTICS / PYTORCH

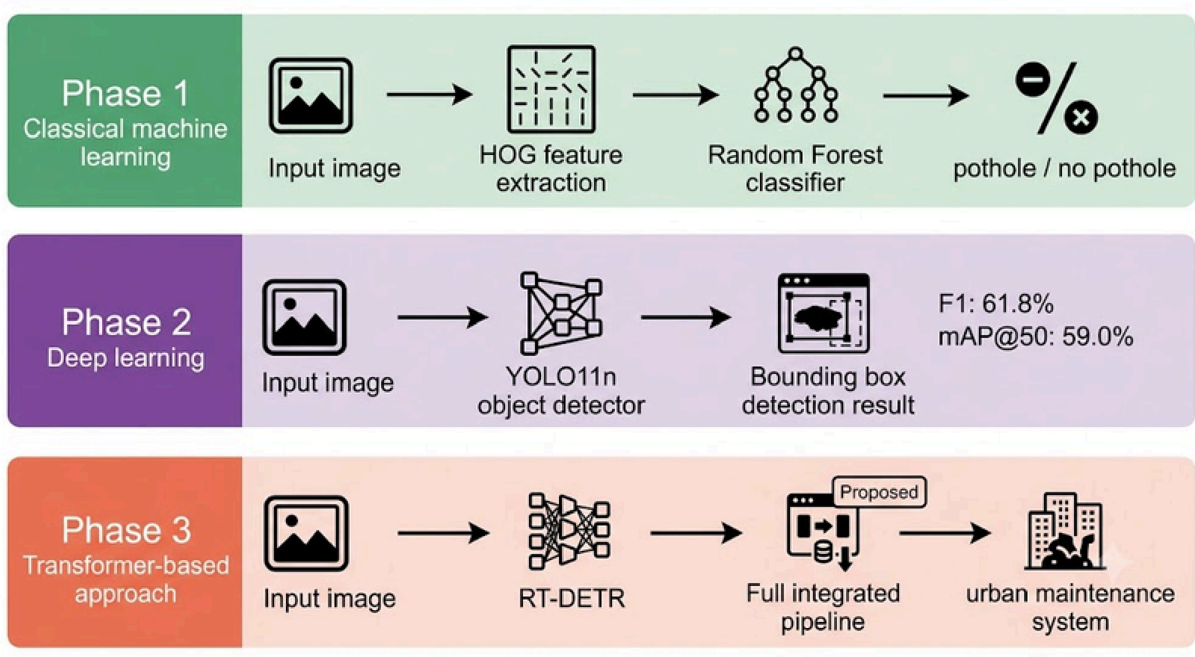
YOLO11n (You Only Look Once Nano) is a real-time object detection model from the YOLO family. The nano variant is optimized for high inference speed and low computational requirements, making it suitable for deployment on resource-constrained devices. Its ability to perform object localization and classification in a single forward pass enables efficient detection of potholes and grates in road images.

UNLIKE YOLO, WHICH USES CONVOLUTIONAL FILTERS WITH A LOCAL RECEPTIVE FIELD, RT-DETR EMPLOYS GLOBAL ATTENTION THAT RELATES ALL REGIONS OF THE IMAGE SIMULTANEOUSLY. IN POTHOLE DETECTION, THIS ALLOWS CONNECTING EDGES WITH DISTANT TEXTURE CHANGES, IMPROVES ACCURACY IN AREAS WITH MULTIPLE NEARBY POTHOLES, AND ELIMINATES THE NEED FOR NMS IN POST-PROCESSING

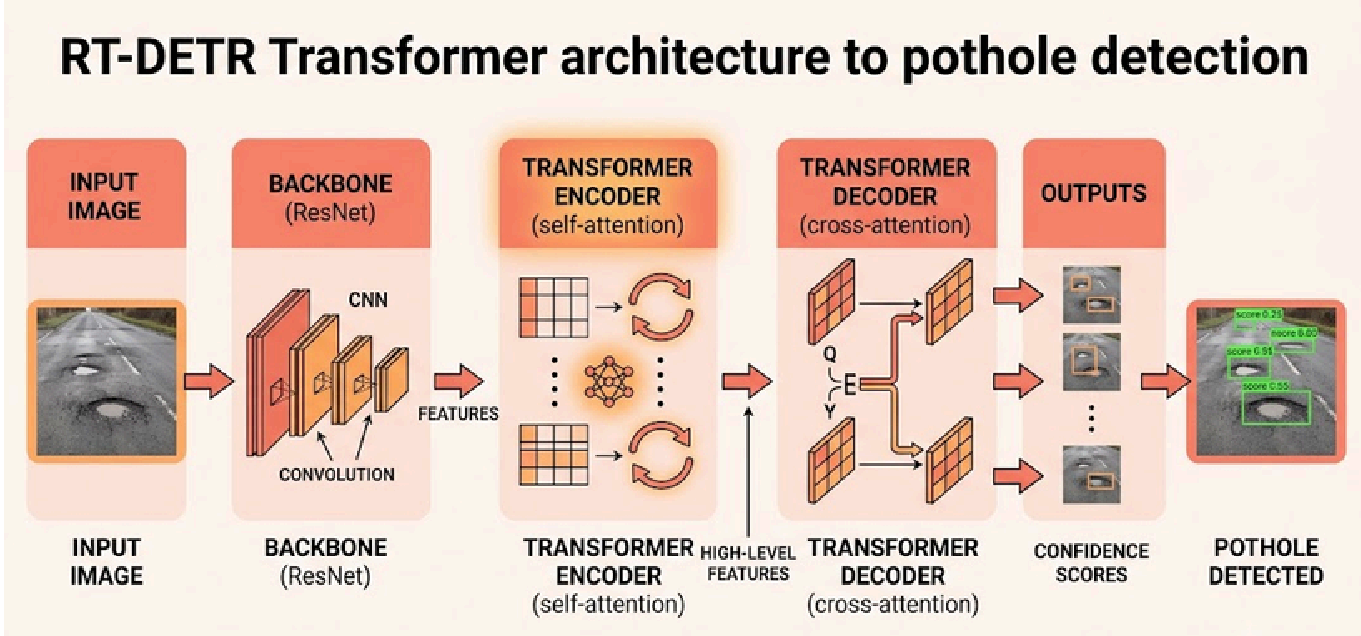
RT-DETR - REAL-TIME DETECTION TRANSFORMER

RT-DETR is a real-time object detection model using Transformers. Its efficient attention mechanism enables accurate detection without complex post-processing like NMS, balancing speed and accuracy—ideal for urban road hazard detection."

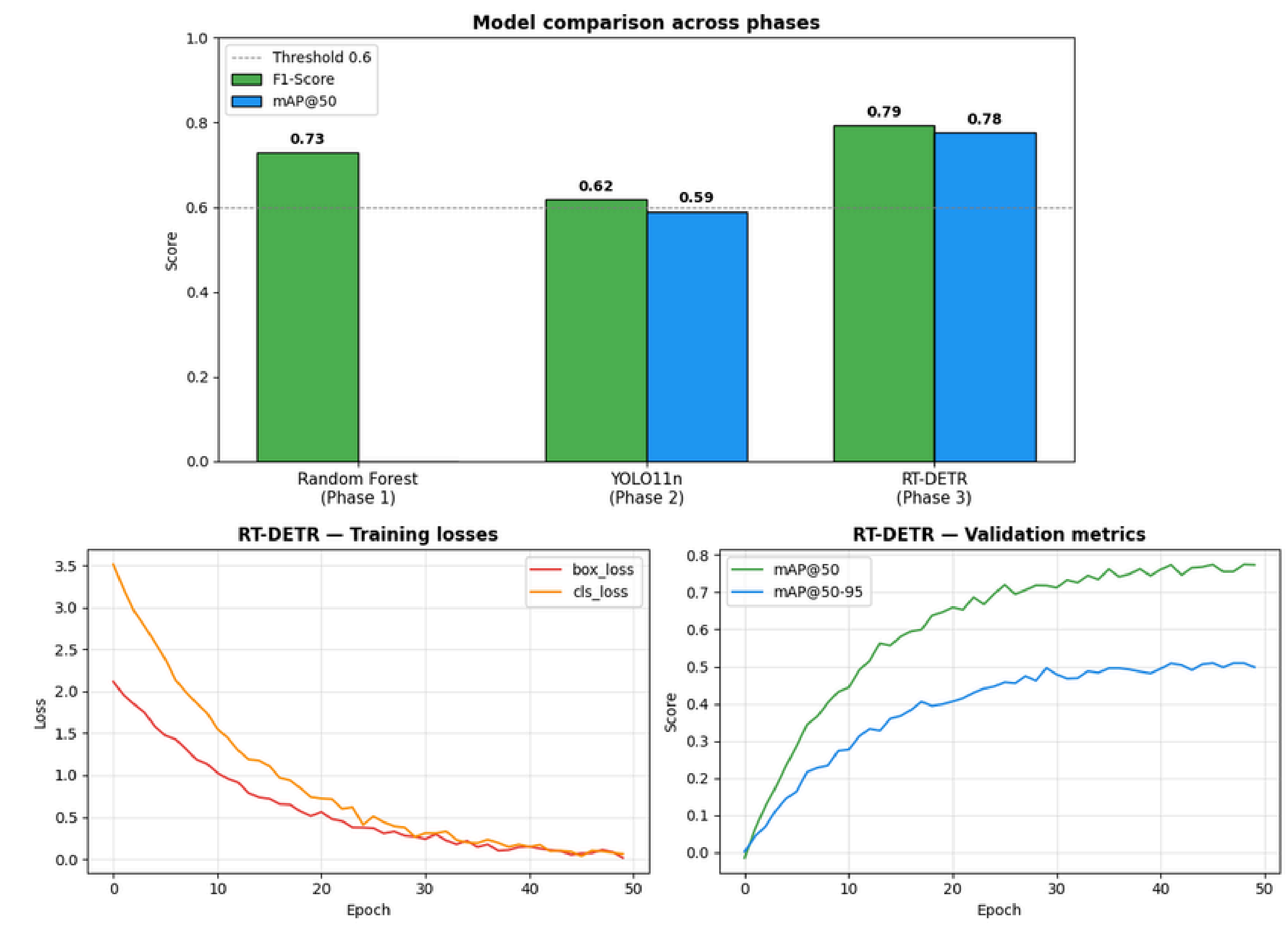
PROJECT PHASES



PIPELINE ARCHITECTURE



QUANTITATIVE RESULTS



F1-SCORE: 79.2% — MAP@50: 77.6%

PREDICTIONS WITH BOUNDING BOXES

ANALYSIS OF FALSE POSITIVES AND FALSE NEGATIVES

- False Positive: shadow or metal grate detected as a pothole due to confusion with dark texture.
- False Negative: pothole with a puddle not detected — the reflection camouflages the defect.

CONCLUSION

The project demonstrated that the progressive application of artificial intelligence to road images enables effective automated pothole detection. From static classification with Random Forest to real-time detection with YOLO11n and the improved accuracy of RT-DETR, each phase enhanced the system's performance. The impact goes beyond the technical: such a tool directly benefits drivers, transport operators, and vulnerable road users, supporting municipalities in prioritizing road maintenance.

Bibliography

- Vaswani, A., et al. (2017). Attention is all you need. arXiv. https://arxiv.org/abs/1706.03762
- Verma, M. (2024). Pothole detection dataset: YOLOv11 optimized. Kaggle. https://www.kaggle.com/datasets/muskanverma24/pothole-detection-dataset-yolov11-optimized
- Ultralytics. (2025). Documentación oficial YOLO11. https://docs.ultralytics.com/es
- Orellana Cusihuaman. (2026). Detección de baches y rejillas — YOLO Fase 2 [Notebook de Google Colab]. https://colab.research.google.com/drive/1CmRNvUGjMQFS29TpNmrMdkrfwHztX4hW